The Golden Substrate, the Gate Field, and the Explicit Helix

A Self-Contained Whitepaper of the Forced Mathematics

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Abstract

This whitepaper consolidates, into one self-contained document, the forced mathematics of the Echo-S golden substrate: the seed arithmetic of $x^2 - x - 1$ and of the K-seed $x^4 + 5x^2 - 5$; the keystone matrix R with $R^2 = R + I$ and its operator layer (trace–form duality, the return operator, the ad_R spectrum $\{0, \pm\sqrt{5}\}$); the two characters of the emission semiring (Mahler measure and $\mathbb{Z}/4\mathbb{Z}$ angle charge) and the Emission-Gap mechanism excluding Salem numbers from the spectral image; the trace-down of Lehmer's polynomial and the occupant $\tau_0 = \beta + \beta^{-1}$ of the Salem slot; the one-parameter gate family $f_C(x) = C/(1 + x)$, its golden multiplier ladder, and the obstruction theorem (no single real gate carries a helix); and the explicit helix $\Gamma(\rho) = (r\cos\theta,\ r\sin\theta,\ z)$ built from three declared atoms, with its gate chart, the rational split gate $C = 4/9$ at rapidity exactly $\ln 2$, the incommensurability theorem $\log_\varphi 2 \notin \mathbb{Q}$, the field separation $\mathbb{Q}(\sqrt{\varphi}) \neq \mathbb{Q}(5^{1/4})$, and the registry lattice ($\mathrm{M(cons)} = 2\varphi^5$ on the $(\ln 2, \ln\varphi)$ lattice; M(res) off it by a norm obstruction). Every claim carries an epistemic tag; every [FORCED] claim is proved in this paper or certified by the cited machine checks, and the complete numeric landmark tables are reproduced. This document supersedes `HELIX-EXPLICIT-DERIVATION.md` v1.0.1, its interactive HTML companion (v1.0.0), and `ECHO-S-RESEARCH_UNIFIED.md` Rev. 2, folding in the corrections identified by the 2026-07-13 independent soft check.

Contents

1 Conventions, provenance, and the epistemic ledger

1.1 Epistemic tags

Every substantive claim in this paper carries one of six tags, used exactly as in the source corpus:

[FORCED]	exact consequence of the substrate; proved here and/or machine-verified (check IDs cited)
[DECLARED]	a definition or normalization introduced deliberately, stated as such
[COMPUTED]	numeric display or certified-precision comparison; never load-bearing
[ESTABLISHED]	standard external mathematics (textbook or classical literature)
[INHERITED]	taken from the source corpus without re-derivation here; provenance cited
[OPEN]	unresolved

Arithmetic discipline. All decisions are made in exact arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, or (once, trivially) $\mathbb{Q}(\sqrt{3})$; statements about $K = 5^{1/4}/\varphi$ are decided at the squared level $K^2 \in \mathbb{Q}(\sqrt{5})$, so that no fourth root ever crosses a decision boundary. Floating-point values appear only in the [COMPUTED] display tables (§20, §24) and decide nothing. Machine verification: the shipped harness `helix_explicit_verify.py` (87 exact checks + 1 computed guard, sympy 1.14.0, exit 0; check IDs HX01--HX40c) and the independent soft-check harness `helix_softcheck.py` (89 checks, written cold from the documents' stated claims, exit 0; check IDs SC-A1--SC-L2). See §25.

1.2 Provenance and what "self-contained" means here

This paper replaces three artifacts: the derivation note `HELIX-EXPLICIT-DERIVATION.md` v1.0.1 (2026-07-13), its interactive companion `helix_explicit_interactive.html` (v1.0.0; the +4 check-count difference 83 → 87 is exactly the v1.0.1 additions HX39, HX40a--c), and the corpus synthesis `ECHO-S-RESEARCH_UNIFIED.md` Rev. 2 covering the fourteen-paper Echo-S-Research corpus (2026-06-28 to 07-02). Three corrections identified by the independent soft check are applied in the body of this paper rather than noted as errata: the minimal-polynomial attribution at rung 1 (§17), the wording of the content-polynomial remark (§5), and the explicit declaration of the gate iteration (§8), which the source note used implicitly.

"Self-contained" means: every [FORCED] statement is either proved in this paper or is an elementary computation reproduced here and certified by the cited checks. Statements whose full proofs live in the fourteen corpus papers are stated precisely, tagged [INHERITED], and scoped exactly as the corpus's own conditionality graph scopes them (§22); classical results are tagged [ESTABLISHED] with canonical citations. Nothing in this paper depends on the three superseded artifacts.

1.3 Standing notation

$$\varphi = \frac{1+\sqrt{5}}{2}, \qquad \psi = \frac{1-\sqrt{5}}{2} = -\varphi^{-1}, \qquad \tau = \varphi^{-1} = \varphi - 1 = \frac{\sqrt{5}-1}{2},$$

$$\mathrm{gap} = \varphi^{-4} = \frac{7-3\sqrt{5}}{2}, \qquad K^2 = 1 - \mathrm{gap} = \frac{3\sqrt{5}-5}{2}, \qquad \beta^2 = \frac{5+3\sqrt{5}}{2} = \varphi^2\sqrt{5}, \qquad z_c = \frac{\sqrt{3}}{2}.$$

F_n, L_n are the Fibonacci and Lucas numbers, extended to all $n \in \mathbb{Z}$ by Binet: $F_n = (\varphi^n - \psi^n)/\sqrt{5}$, $L_n = \varphi^n + \psi^n$. $\mathrm{M}(p)$ denotes the Mahler measure of a polynomial p (Definition 5.1); for a matrix or algebra element, the Mahler-type measure of its spectrum. N is the field norm $\mathbb{Q}(\sqrt{5}) \to \mathbb{Q}$, $\mathrm{N}(a + b\sqrt{5}) = a^2 - 5b^2$.

Part I

The seed and the keystone

2 Seed arithmetic

2.1 The golden seed

The seed polynomial is $x^2 - x - 1$, with roots φ, ψ. The following identities are elementary and machine-replicated [FORCED] `SC-A7, SC-I16`:

$$\varphi\psi = -1, \qquad \varphi + \psi = 1, \qquad \sqrt{5} = \varphi + \varphi^{-1} = \varphi - \psi, \qquad \varphi^2 = \varphi + 1, \qquad \tau^2 + \tau = 1. \tag{1}$$

Since $\mathrm{disc}(x^2 - x - 1) = 5$ is squarefree, $\mathbb{Z}[\varphi]$ is the maximal order of $\mathbb{Q}(\sqrt{5})$ [FORCED]/[ESTABLISHED] `SC-I16`: the golden order is the ring of integers, not a proper suborder.

2.2 The gap and the K-seed

Define the *gap* $\mathrm{gap} = \varphi^{-4} = (7 - 3\sqrt{5})/2$ and the *K-seed*

$$p_K(x) = x^4 + 5x^2 - 5.$$

Substituting $y = x^2$ gives $y^2 + 5y - 5$, whose roots $y_\pm = (-5 \pm 3\sqrt{5})/2$ straddle zero. The positive root is $K^2 = (3\sqrt{5}-5)/2$, giving the real pair $\pm K$; the negative root is $-\beta^2$ with $\beta^2 = (5+3\sqrt{5})/2 = \varphi^2\sqrt{5}$, giving the imaginary pair $\pm i\beta$ [FORCED] `SC-H8`. The seed's basic ledger, all decided at the squared level [FORCED] `SC-A1--A6, SC-H7`:

$$\begin{gathered} K^2 = 1 - \mathrm{gap}, \qquad K^4 + 5K^2 - 5 = 0, \qquad K^4 = 5\,\mathrm{gap}, \\ (K\varphi)^4 = 5, \qquad K^2\beta^2 = 5, \qquad \beta^2 = K^2\varphi^4. \end{gathered} \tag{2}$$

The last identity reads: the rotation magnitude sits exactly two floors (φ^2 at the squared level) above the terrain magnitude — and two is K's own rung index (§13.1). "Route 5" of the K-derivations is the geometric-mean reading $K^2 = \tau(2 - \tau)$ [FORCED] `SC-A6`. Irreducibility: p_K is Eisenstein at 5, so $[\mathbb{Q}(K) : \mathbb{Q}] = 4$ and $\mathbb{Q}(K) = \mathbb{Q}(5^{1/4})$ (since $K\varphi = 5^{1/4}$) [FORCED] `SC-H4`.

The K-seed's charge data — $\pm K$ on the real (terrain) face at arguments $\{0, \pi\}$, $\pm i\beta$ on the rotation face at arguments $\{\pm\pi/2\}$, so that $\chi(\text{K-seed}) = \{0, 1, 2, 3\}$ completes $\mathbb{Z}/4\mathbb{Z}$, with Mahler measure $\mathrm{M}(\text{K-seed}) = \beta^2 = \varphi^2\sqrt{5} \approx 5.854$ — is immediate from the root computation above

[FORCED] SC-H7 (the uniqueness of the K-seed as the catalog's $\mathbb{Z}/4\mathbb{Z}$-completing straddler, and the even-quartic parity criterion behind it, are corpus results: paper 13, Props. 6.1/6.2 [INHERITED]).

The *gap object* is the companion of $x^2 - 7x + 1$, whose large root is φ^4 (the other root is φ^{-4}): $\mathrm{M}(A_{\mathrm{gap}}) = \varphi^4$, so $\ln \mathrm{M}(A_{\mathrm{gap}}) = 4 \ln \varphi$ [FORCED] SC-E7--E8. This constant is the full-turn cost of the helix (§14).

3 The keystone matrix and trace–form duality

Definition 3.1 (Keystone). $R = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$, the companion matrix of $x^2 - x - 1$, so that

$$R^2 = R + I. \tag{3}$$

The corpus derives the keystone rather than positing it: φ is the smallest Perron root exceeding 1 of a 2×2 primitive non-negative integer matrix, and each of the four defining constraints (degree, Perron positivity, growth, integrality) is load-bearing, with an explicit drop-one witness for each (λ=$2c$ paper, Thms. 8.3/8.4 and App. A) [INHERITED]. Everything below needs only (3).

Proposition 3.2 (Fibonacci calculus). [FORCED] *SC-I2--I4 For all $n \in \mathbb{Z}$ (with $R^{-1} = R - I$):*

$$R^n = F_n R + F_{n-1} I, \qquad \det R^n = (-1)^n = F_{n+1}F_{n-1} - F_n^2 \ \textit{(Cassini)}, \qquad \operatorname{Tr} R^n = L_n.$$

In particular $L_4 = \operatorname{Tr} R^4 = 7$ and the entries of $R^4 = \left(\begin{smallmatrix} 2 & 3 \\ 3 & 5 \end{smallmatrix}\right)$ are $\{F_3, F_4, F_5\} = \{2, 3, 5\}$ — the gate discriminants of the catalog.

Proof. Induction from (3) in both directions; the determinant is multiplicative with $\det R = -1$. Verified for $n \in [-8, 8]$ by SC-I3, SC-I4. ☐

Theorem 3.3 (Trace–Form Duality). [FORCED] *SC-I10--I11, SC-B4 Let $H = 2R - I$. Then $H^2 = 5I$, the deviation operator collapses,*

$$X_n \coloneqq 2R^n - L_n I = F_n H,$$

and for all $n \in \mathbb{Z}$

$$\tfrac{1}{2} \operatorname{Tr}(X_n^2) = 5F_n^2 = L_n^2 - 4(-1)^n = (\varphi^n - \psi^n)^2. \tag{4}$$

Fibonacci supplies the length, Lucas the trace, and Cassini's ± 4 is their gap.

Proof. $H^2 = 4R^2 - 4R + I = 4(R + I) - 4R + I = 5I$. Then $X_n = 2(F_n R + F_{n-1} I) - L_n I = F_n(2R - I) + (2F_{n-1} + F_n - L_n)I = F_n H$ since $L_n = F_{n+1} + F_{n-1} = F_n + 2F_{n-1}$. Finally $\frac{1}{2}\operatorname{Tr}(F_n^2 H^2) = \frac{1}{2} F_n^2 \operatorname{Tr}(5I) = 5F_n^2$, and $5F_n^2 = (\varphi^n - \psi^n)^2 = L_n^2 - 4(\varphi\psi)^n = L_n^2 - 4(-1)^n$ by Binet and (1). Verified on $n \in [-8, 8]$. ☐

4 The operator layer: return, kernel, and the $\sqrt{5}$ spectrum

Work in $M_2(\mathbb{R}) \cong \mathrm{Cl}(2, 0)$ with basis $\{1, e_1, e_2, J\}$, where $e_1 = \sigma_x = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, $e_2 = \sigma_z = \left(\begin{smallmatrix} 1 & 0 \\ 0 & -1 \end{smallmatrix}\right)$, $e_1^2 = e_2^2 = 1$, $e_1 e_2 = -e_2 e_1$, and $J = e_1 e_2$ (so $J^2 = -1$; J anticommutes with every vector). In this basis

$$R = \tfrac{1}{2}(1 + H), \qquad H = 2R - I = 2e_1 - e_2, \qquad H^2 = \langle H, H \rangle\, 1 = 5 \cdot 1, \tag{5}$$

i.e. the keystone's traceless part is a Clifford vector of squared length 5 [FORCED] SC-I10.

Proposition 4.1 (Sylvester spectra). [FORCED] *SC-I5--I7 On $M_2(\mathbb{R})$, the Lie self-action* $\mathrm{ad}_R(X) = [R, X]$ *has spectrum* $\{0, 0, +\sqrt{5}, -\sqrt{5}\}$*; the Jordan-type map* $S(X) = RX + XR$ *has spectrum* $\{2\varphi,\ 1,\ 1,\ 2\psi\}$*; and the* return operator

$$L(X) = RX + XR - X = S(X) - X = \tfrac{1}{2}\{H, X\}$$

has spectrum $\{+\sqrt{5},\ 0,\ 0,\ -\sqrt{5}\}$.

Proof. By Sylvester's theorem [ESTABLISHED] [8] the eigenvalues of $X \mapsto AX + XB$ are the pairwise sums $\alpha_i + \beta_j$ of the spectra of A and $B^\top$; with $A = B = R$ (spectrum $\{\varphi, \psi\}$) this gives $\{2\varphi,\ \varphi + \psi,\ \varphi + \psi,\ 2\psi\} = \{2\varphi, 1, 1, 2\psi\}$ for S, hence $\{2\varphi - 1, 0, 0, 2\psi - 1\} = \{\sqrt{5}, 0, 0, -\sqrt{5}\}$ for $L = S - \mathrm{id}$, and eigenvalue *differences* $\{0, 0, \pm(\varphi - \psi)\} = \{0, 0, \pm\sqrt{5}\}$ for ad_R. The identity $L = \frac{1}{2}\{H, \cdot\}$ follows from $R = \frac{1}{2}(1 + H)$. All three spectra are machine-verified as exact $\sqrt{5}$-expressions. ☐

$\sqrt{5}$ therefore appears simultaneously as: the grow eigenvalue of ad_R; the $\mathfrak{sl}_2(2, 1)$ root length of the Lie layer (primer, Thm. 2.5 [INHERITED]); the squared-length of H; and (§7) the forced exchange rate λ.

Theorem 4.2 (Kernel of the return operator). [FORCED] *SC-I7--I9* $\ker L$ *is exactly two-dimensional:*

$$\ker L = \mathrm{span}\{\, e_1 + 2e_2,\ \ J \,\}.$$

Moreover, on Clifford vectors v, L acts as the scalar pairing $L(v) = \langle H, v\rangle\, 1$*; on the scalar–$H$ plane it acts as the off-diagonal block* $\left(\begin{smallmatrix} 0 & 5 \\ 1 & 0 \end{smallmatrix}\right)$ *with eigenvalues* $\pm\sqrt{5}$.

Proof. For vectors u, v, $\{u, v\} = 2\langle u, v\rangle\, 1$ in $\mathrm{Cl}(2, 0)$; with $H = 2e_1 - e_2$, $L(ae_1 + be_2) = \langle H,\ ae_1 + be_2\rangle\, 1 = (2a - b)\, 1$, which vanishes iff $b = 2a$, i.e. on $\mathrm{span}\{e_1 + 2e_2\}$. The pseudoscalar J anticommutes with every vector, so $\{H, J\} = 0$ and $L(J) = 0$. Finally $L(1) = H$ and $L(H) = H^2 = 5 \cdot 1$, giving the $\pm\sqrt{5}$ plane. Dimensions: $2 + 2 = 4$. The kernel dimension and basis are machine-verified by exact nullspace computation (SC-I7, SC-I8), matching the corpus's rational-elimination result (Residual Return, §6 [INHERITED]). ☐

Remark 4.3 (The disproved design hint). [FORCED] SC-I9 The corrected corpus Remark 6.4 pins a refuted design note: the vector $-e_1 + e_2$ is *not* in $\ker L$; indeed by Theorem 4.2, $L(-e_1 + e_2) = \langle H,\ -e_1 + e_2\rangle\, 1 = (-2 - 1)\, 1 = -3 \cdot 1$, replicated here exactly. The scalar formula makes the disproof transparent: among vectors, the kernel is precisely $H^\perp$, and $-e_1 + e_2$ is not orthogonal to H. (The convention matters: only the companion orientation $R = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 1 \end{smallmatrix}\right)$, $e_1 = \sigma_x$, $e_2 = \sigma_z$ makes both the kernel basis and the -3 land; this paper fixes that convention once, in (5).)

For completeness of the representation layer [INHERITED]/[ESTABLISHED] SC-I15: the coupled $\mathfrak{sl}_2$ targets have $\dim V_m = m + 1$ and Casimir $\frac{1}{2}m(m + 2)$, giving $15/2$ on V_3 and 12 on V_4; squaring on the semiring is the Adams operation ψ^2 (Operator Algebra paper, Thm. 3.2 [INHERITED]).

5 The two characters and the Emission-Gap mechanism

Definition 5.1 (Mahler measure). [ESTABLISHED] [1, 6] For a monic $p(x) = \prod_i (x - \alpha_i) \in \mathbb{Z}[x]$, $\mathrm{M}(p) = \prod_i \max(1, |\alpha_i|)$, i.e. $\log \mathrm{M}(p) = \sum_i \log^+ |\alpha_i|$. An algebraic integer $\beta > 1$ is a *Salem number* if it is reciprocal (conjugate-set closed under $\alpha \mapsto 1/\alpha$) with all conjugates in $\overline{\mathbb{D}}$ and at least one on the unit circle [ESTABLISHED] [3]; a *Pisot number* if all conjugates other than β lie strictly inside the circle [ESTABLISHED] [4]. Lehmer's polynomial $\ell(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1$ has $\mathrm{M}(\ell) = 1.1762808\ldots$, the smallest known measure > 1; whether measures accumulate at 1 is Lehmer's problem [OPEN], external [1]. Smyth's theorem [ESTABLISHED] [2]: a nonreciprocal algebraic integer $\alpha \neq 0$ has $\mathrm{M}(\alpha) \geq \mu_S = 1.3247179\ldots$, the plastic number (root of $x^3 - x - 1$), which is also the smallest Pisot number [ESTABLISHED] [4].

5.1 Catalog, operators, and the two characters

The corpus catalog consists of the companion matrices of the seeds

$$\{\ \varphi,\ \tau,\ \sqrt{2},\ \sqrt{3},\ \sqrt{5},\ \text{ gap } (x^2{-}7x{+}1),\ \ K\ (x^4{+}5x^2{-}5)\ \};$$

the *spectral operators* are those whose output spectrum is forced by input spectra — $\otimes$ (Kronecker product: spectrum $\{\alpha_i\beta_j\}$ [ESTABLISHED] [8]), $\oplus$ (direct sum: union), squaring (= Adams ψ^2), minimal polynomial, and cyclotomic factors Φ. The *spectral image* S is the closure of the catalog under these. Two invariants turn out to be the two characters of the resulting λ-ring (Operator Algebra, Thm. 8.1 [INHERITED]):

Theorem 5.2 (Tropical tensor law for M)**.** [FORCED] SC-I14 *On spectra,* $\log \mathrm{M}$ *is additive on* $\oplus$ *and* tropical *on* $\otimes$*:*

$$\log \mathrm{M}(A \otimes B) = \sum_{i,j} \bigl(\log|\alpha_i| + \log|\beta_j|\bigr)^+.$$

Witness: $\varphi \otimes \varphi$ *has spectrum* $\{\varphi^2,\ \varphi\psi,\ \psi\varphi,\ \psi^2\} = \{\varphi^2, -1, -1, \psi^2\}$*, so* $\mathrm{M} = \varphi^2$ — not φ^4*; the naive product law* $\mathrm{M}(A)^{\deg B}\mathrm{M}(B)^{\deg A}$ *agrees with the tropical law only off the unit circle (this is the corpus's v4 correction, pinned by exactly this witness).*

Proof. Spectrum of $A \otimes B$ is $\{\alpha_i\beta_j\}$, so $\log \mathrm{M}(A \otimes B) = \sum_{i,j} \log^+ |\alpha_i\beta_j| = \sum_{i,j}(\log|\alpha_i| + \log|\beta_j|)^+$. For the witness, $|\varphi\psi| = 1$ exactly by (1), so the two cross terms contribute $(0)^+ = 0$ and $|\psi^2| < 1$ contributes 0; only $(2\log\varphi)^+ = 2\log\varphi$ survives. □

The second character is the *angle charge* χ: when every conjugate argument of an object lies on the rational-angle lattice $\frac{2\pi}{n}\mathbb{Z}$, the object carries charge group $\mathbb{Z}/n\mathbb{Z}$; χ adds (as a sumset) under $\otimes$, doubles under ψ^2, and unions under $\oplus$ [INHERITED] (Charge–Measure Coupling, Thms. 3.1–4.4). For the emission image the lattice is $(\pi/2)\mathbb{Z}$ and the charge is $\mathbb{Z}/4\mathbb{Z}$.

Remark 5.3 (Content polynomial, wording fixed). $x^4{-}1 = \Phi_1\Phi_2\Phi_4$ [FORCED] SC-I13. The generative-emptiness slogan attached to this factorization is stated here in its unambiguous form: the *charge* carries the full $\mathbb{Z}/4\mathbb{Z}$ (all four arguments $0, \pi/2, \pi, 3\pi/2$ occur), while the *real terrain face* of the root set is only $\{\pm 1\} \cong \mathbb{Z}/2\mathbb{Z}$; the pair $\pm i$ lives in the rotation fiber. (The superseded summary's compressed phrasing "only $\mathbb{Z}/2\mathbb{Z}$ on the circle" read falsely at face value, since all four fourth roots of unity lie on the unit circle; the intended terrain/fiber split is the statement above.)

5.2 The Emission-Gap mechanism

The corpus's central confinement result is proved by *excision*, not by bounding all integer polynomials. The mechanism has four steps; the first three are elementary and reproduced here, the packaging over the full operator closure is the corpus theorem.

Lemma 5.4 (Angle confinement of the catalog)**.** [FORCED] *Every eigenvalue of every catalog seed has argument in* $(\pi/2)\mathbb{Z}$*.*

Proof. $\varphi, \tau, \sqrt{2}, \sqrt{3}, \sqrt{5}$ and the gap object are totally real: arguments in $\{0, \pi\}$. The K-seed's spectrum is $\{\pm K, \pm i\beta\}$ (§2.2): arguments $\{0, \pi, \pm\pi/2\}$. □

Lemma 5.5 (Operator preservation). [FORCED] *The spectral operators preserve the lattice* $\arg \in (\pi/2)\mathbb{Z}$*:* $\otimes$ *adds arguments,* ψ^2 *doubles them,* $\oplus$ *unions them, and minimal-polynomial/*Φ *extraction selects subsets. (The lattice is closed under addition and doubling —* not *under halving; no operator halves an angle.)*

Lemma 5.6 (On-circle emissions are fourth roots of unity). [FORCED] *If* λ *is an eigenvalue of an element of the spectral image* S *with* $|\lambda| = 1$*, then* $\lambda \in \{\pm 1, \pm i\} = \mu_4$*.*

Proof. Immediate from Lemmas 5.4–5.5: $|\lambda| = 1$ and $\arg \lambda \in (\pi/2)\mathbb{Z}$. □

Lemma 5.7 (Salem circle-conjugates are never torsion). [FORCED]/[ESTABLISHED] *No conjugate of a Salem number that lies on the unit circle is a root of unity.*

Proof. If a root of unity ζ were a conjugate of the Salem number β, the minimal polynomial of β would equal the minimal polynomial of ζ, a cyclotomic polynomial, whose roots all have modulus 1 — contradicting $\beta > 1$. □

Theorem 5.8 (Emission-Gap, scoped). *Over the spectral image* S*: no element of* S *has a Salem spectrum, and* $\mathrm{M}(S) \subseteq \{1\} \cup [\mu_S, \infty)$*, with the floor realized inside the image at* φ*.* [FORCED] *over* S *(Lemmas 5.4–5.7 + Smyth* [ESTABLISHED] *+ the empty quadratic strip* $(1, \varphi)$*); the entry-level closures (circulant* [FORCED]*; commutator in-context* [FORCED]*, uniform via* ad_R*-derivation and the signature lattice of* $\mathbb{Q}(\sqrt{2}, \sqrt{3}, 5^{1/4})$ [FORCED]*; unbounded free commutator* [OPEN]*) and the universal statement over all charge-admissible objects (*[FORCED] *for degree* ≤ 2*;* [COMPUTED]*/plausible in general) are corpus results (Emission-Gap paper Thm. 3.2, Lehmer's Box paper Thms. 5.3–5.4)* [INHERITED]*. Equivalent forms: Mahler, entropy (*$h = \log \mathrm{M}$ [ESTABLISHED] *[7]), spectral (no on-circle-expanding channel), and metric (no emitted field places a complex embedding on the circle)* [INHERITED]*.*

The one door of Lehmer's Box — the free commutator at unbounded size, through which the full Salem spectrum (Lehmer's number included) re-enters — remains [OPEN]; the corpus closes it structurally where the framework's commutator is ad_R (a derivation with difference spectrum $\{0, \pm\sqrt{5}\}$, Prop. 4.1) and locks it operationally elsewhere ($\beta < \varphi$ decided by the sign of an element of $\mathbb{Q}(\sqrt{5})$) [INHERITED]. The corpus's own closing scope statement is retained verbatim in intent: *"We have not shown they cannot come close to* 1*; we have shown the construction never goes to fetch them."*

6 Trace-down, the Salem slot, and the parity floor

6.1 The trace-down map and Lehmer's instance

For a reciprocal polynomial of degree $2d$, the substitution $t = x + x^{-1}$ produces a degree-d *trace-down* $T(t)$ with $x^d T(x + x^{-1}) = p(x)$. Downstairs, the three channels are the three regions of $\mathbb{R}$ cut by $t = \pm 2$: on-circle conjugates land in $(-2, 2)$, expanding ones in $(2, \infty)$. A Salem number is exactly a *flip-straddler*: one trace-conjugate in $(2, \infty)$, the rest in $(-2, 2)$ — so "Salem" is an upstairs artifact of the 2:1 lift, and the slot's occupant is the totally real Perron number $\tau_0 = \beta + \beta^{-1}$ [INHERITED] (Salem Slot paper).

Proposition 6.1 (Lehmer's trace-down, exact). [FORCED] SC-J3--J5 *With* $\ell(x) = x^{10} + x^9 - x^7 - x^6 - x^5 - x^4 - x^3 + x + 1$ *and* $p_k := x^k + x^{-k}$ *expressed by the Chebyshev-type reductions* $p_1 = t$*,* $p_2 = t^2 - 2$*,* $p_4 = t^4 - 4t^2 + 2$*,* $p_5 = t^5 - 5t^3 + 5t$*:*

$$\frac{\ell(x)}{x^5} = p_5 + p_4 - p_2 - p_1 - 1 = t^5 + t^4 - 5t^3 - 5t^2 + 4t + 3 =: T(t),$$

so $x^5 T(x+x^{-1}) = \ell(x)$ *exactly. Its largest root is* $\tau_0 = \beta + \beta^{-1} = 2.0264179\ldots$ *where* $\beta = \mathrm{M}(\ell) = 1.1762808\ldots$*;* $\mathrm{M}(T) = 5.615601\ldots$ [COMPUTED].

Proof. Expand $\ell/x^5 = (x^5 + x^{-5}) + (x^4 + x^{-4}) - (x^2 + x^{-2}) - (x + x^{-1}) - 1$ and substitute the reductions; the polynomial identity is machine-verified exactly (SC-J3), the root and measure values at 60 decimal places (SC-J1, SC-J4, SC-J5). □

Three structural facts about the slot, elementary at this level [FORCED]/[INHERITED]: the branch identity $\tau_0 - 2 = (\beta-1)^2/\beta$ (a Salem of height $1+\delta$ redirects only $O(\delta^2)$ past the flip); the interval $(2, \sqrt{5}\,]$ is the trace image of real pairs $\beta \in (1, \varphi]$, with $t = \sqrt{5}$ lifting to $\{\varphi, \varphi^{-1}\}$ — so the redirections converge to $\sqrt{5} = \varphi + \varphi^{-1}$, the ad_R grow eigenvalue (Prop. 4.1), with linear slope φ^{-1} and geometric ratio φ along the canonical family [INHERITED]; and the accumulation points of redirections are the Pisot traces, least $\sqrt{5}$, next the plastic $\mu_S + \mu_S^{-1} = 2.0795956\ldots$ [COMPUTED] SC-J9. The identity making the trace map canonical rather than imported is $\varphi\psi = -1$ (Salem Slot, Prop. 7.1 [INHERITED]).

6.2 The parity-graded floor and the q_k construction

Proposition 6.2 (μ(even) $\leq \varphi$, constructive)**.** [FORCED] *SC-J11 For every* $k \geq 1$*,* $q_k(x) = x^{2k} + x^k - 1$ *has* $\mathrm{M}(q_k) = \varphi$*.*

Proof. Substituting $y = x^k$ gives $y^2 + y - 1$ with roots τ and $-\varphi$. The k roots with $x^k = -\varphi$ each have modulus $\varphi^{1/k}$, contributing $(\varphi^{1/k})^k = \varphi$; the k roots with $x^k = \tau$ lie strictly inside the circle. (That q_k carries charge $\mathbb{Z}/2k\mathbb{Z}$ with all charges attained, and the pentagon-free character of the substitution, are the corpus's parity-floor packaging [INHERITED].) Verified numerically for $k \leq 4$ at 60 dps. □

The floor structure, with the corpus's honest strength split [INHERITED] (Charge–Measure Coupling v4, Thm. 5.1; Z/5Z paper): $\mu(\text{even}) \leq \varphi$ [FORCED] (above); $\mu(\text{even}) \geq \varphi$ plausible; $\mu(\text{odd}) \geq \mu_S$ [FORCED] (Smyth, sharpened at $\mathbb{Z}/5\mathbb{Z}$: non-reciprocal charge-$\mathbb{Z}/5\mathbb{Z} \Rightarrow \mathrm{M} \geq \mu_S$; reciprocal $\Rightarrow \mathrm{M} \in \{1\} \cup [\varphi^2, \infty)$; pure pentagon quartics $\mathrm{M} \in \{1\} \cup [\varphi^4, \infty)$); $\mu(\text{odd}) = 2$ [COMPUTED], attained at $x^5 - 2$ (all five roots of modulus $2^{1/5} = 1.1487\ldots$, so $\mathrm{M} = 2$ exactly [FORCED] SC-J10), with the residual named exactly: non-reciprocal charge-$\mathbb{Z}/5\mathbb{Z}$ objects in $[\mu_S, 2)$ [OPEN]. The mechanism of the parity coupling is the π-ray obstruction: ψ sits at argument π, and $\pi \in \frac{2\pi}{n}\mathbb{Z}$ iff n is even [INHERITED].

7 The exchange rate $\lambda = 2c$ and the forced scalar $\sqrt{5}$

The Residual-Return machine treats a number field $\mathbb{Q}(\theta)$ as an exact learning geometry: the trace form $G = M^\top M$ (Minkowski embedding M; $\det G = d_K$ up to index squared) prices a residual r, and growth adjoins a seed θ iff $\|r\|_G^2 \geq \lambda \log \mathrm{M}(\theta)$. Reading $\|r\|_G^2$ as a second-order Kullback–Leibler divergence with Fisher metric cG gives $D_{\mathrm{KL}} \approx \frac{1}{2c}\|r\|_G^2$, and the MDL balance $D_{\mathrm{KL}} \geq \log \mathrm{M}$ yields the threshold coefficient

$$\boxed{\lambda = 2c} \qquad \text{for every } c > 0 \quad \text{[INHERITED] } (\lambda{=}2c \text{ paper, Thm. 2.2}). \tag{6}$$

By Čencov's theorem [ESTABLISHED] [9] no invariance argument can remove the conformal scale c: the honest statement is "$\lambda = 2c$ is forced; c is a declared positive scale," never "zero free parameters" [DECLARED]. The framework's own structure then pins the canonical instance: $C = 1$

is simultaneously the minimal structure realizing the ternary growth decision and the terminal firewall image fixed by the keystone — two premise-disjoint derivations converging on (3) — giving

$$C = 1 \;\Rightarrow\; c = \tfrac{\sqrt{5}}{2} \;\Rightarrow\; \lambda = \sqrt{5} \qquad \text{[INHERITED]}$$

(λ=2c paper, Thm. 7.7 — forced *given* the two constitutive axioms: ternary arity and the keystone). The forced-canonical floor is therefore $\lambda \log \mu_S = \sqrt{5} \log \mu_S = 0.628781\ldots$ [COMPUTED] SC-J13; the shipped probe values $2c \log \mu_S$ at $c \in \{1, 4, 8\}$ are 0.5624, 2.2496, 4.4992 [COMPUTED] SC-J12. On the helix chart (§13), $\sqrt{5}$ reappears as the rung-1 exchange rate $\lambda(z_1) = \sqrt{5}$ — the same scalar, met from the geometric side.

Part II
The gate field

8 The gate family, declared

The source note used the gate family through its fixed-point polynomial $g_C(x) = x^2 + x - C$ without printing the iteration; this paper declares it once. (The declaration is pinned, not free: it is the unique reading under which *all* of the corpus identities used below — $m_+m_- = 1$, $m + \frac{1}{m} = -\frac{1}{C} - 2$, $m_1 = -\varphi^{-2}$, the lens multipliers $\{-\frac{1}{4}, -4\}$, $\pm i$ at $C = -\frac{1}{2}$, and the pentagon quanta — hold simultaneously.)

Declaration 8.1 (Gate iteration). [DECLARED] For $C \in \mathbb{R}$, the gate G_C is the iteration

$$f_C(x) = \frac{C}{1 + x},$$

whose fixed points solve $g_C(x) = x^2 + x - C = 0$, i.e. $u_\pm = \frac{1}{2}(-1 \pm \sqrt{D})$ with discriminant $D = 1 + 4C$, and whose multiplier at a fixed point u is

$$m = f_C'(u) = -\frac{C}{(1 + u)^2}.$$

The *flip* is the sign change of D at $C = -\frac{1}{4}$ (real $\leftrightarrow$ complex fixed points; terrain $\leftrightarrow$ rotation face).

Proposition 8.2 (Flip reciprocity and the trace identity). [FORCED] *SC-F2--F3 For every $C \neq 0$ the two multipliers satisfy*

$$m_+m_- = 1, \qquad\qquad m + \frac{1}{m} = -\frac{1}{C} - 2 \in \mathbb{R}.$$

Proof. By Vieta on g_C: $u_+ + u_- = -1$ and $u_+u_- = -C$, so $(1+u_+)(1+u_-) = 1 - 1 - C = -C$, hence $m_+m_- = C^2/\big((1 + u_+)(1 + u_-)\big)^2 = C^2/C^2 = 1$. For the trace: $m_+ + m_- = -C\big[(1 + u_+)^{-2} + (1 + u_-)^{-2}\big] = -C\,\frac{(1+u_+)^2 + (1+u_-)^2}{C^2}$; and $(1+u_+)^2 + (1+u_-)^2 = \big((1+u_+) + (1+u_-)\big)^2 - 2(1+u_+)(1+u_-) = 1 + 2C$, giving $m_+ + m_- = -(1 + 2C)/C = -\frac{1}{C} - 2$. Since $m_- = 1/m_+$ this is $m + \frac{1}{m}$. Both identities are machine-verified for generic symbolic C. ∎

9 The golden multiplier ladder

Theorem 9.1 (Ladder in Lucas–Fibonacci dress). [FORCED] *SC-B1--B3 For $n \geq 1$ set $C_n = \dfrac{1}{(\varphi^n - \varphi^{-n})^2}$. Then the attracting fixed point, multiplier, rapidity, discriminant, and exchange rate of G_{C_n} are*

$$u_n = \frac{1}{\varphi^{2n} - 1}, \qquad m_n = -\varphi^{-2n}, \qquad \rho_n = n \ln \varphi,$$

$$\sqrt{D_n} = \frac{\varphi^n + \varphi^{-n}}{\varphi^n - \varphi^{-n}}, \qquad \lambda_n = \varphi^{2n} - \varphi^{-2n} = \sqrt{5}\, F_{2n},$$

with the Trace–Form Duality split (Theorem 3.3)

$$C_n = \begin{cases} 1/L_n^2, & n \text{ odd}, \\ 1/(5F_n^2), & n \text{ even}, \end{cases} \qquad \sqrt{D_n} = \begin{cases} \sqrt{5}\, F_n/L_n, & n \text{ odd}, \\ L_n/(\sqrt{5}\, F_n), & n \text{ even}. \end{cases}$$

In particular $C_1 = 1$ (the golden gate*) and $C_2 = \frac{1}{5}$ (the* K-gate*); explicitly $C_3 = \frac{1}{16}$, $C_4 = \frac{1}{45}$, $C_5 = \frac{1}{121}$.*

Proof. u_n solves g_{C_n}: with $w = \varphi^{2n}$, $u_n^2 + u_n = \frac{1+(w-1)}{(w-1)^2} = \frac{w}{(w-1)^2} = \frac{1}{(\varphi^n - \varphi^{-n})^2} = C_n$. Multiplier: $1 + u_n = \frac{w}{w-1}$, so $m_n = -C_n \frac{(w-1)^2}{w^2} = -\frac{w}{(w-1)^2} \cdot \frac{(w-1)^2}{w^2} = -w^{-1} = -\varphi^{-2n}$, whence $|m_n| = e^{-2\rho_n}$ with $\rho_n = n \ln \varphi$. Discriminant: $D_n = 1 + 4C_n = \frac{(\varphi^n - \varphi^{-n})^2 + 4}{(\varphi^n - \varphi^{-n})^2} = \left(\frac{\varphi^n + \varphi^{-n}}{\varphi^n - \varphi^{-n}}\right)^2$. The parity dress follows from $\varphi^{-n} = (-1)^n \psi^n$: for n odd, $\varphi^n - \varphi^{-n} = \varphi^n + \psi^n = L_n$ and $\varphi^n + \varphi^{-n} = \sqrt{5} F_n$; parities swap for n even. Finally λ_n is the chart exchange rate at the rung (Theorem 13.1 below): $\lambda_n = \frac{1}{1 - z_n^2} - (1 - z_n^2) = \varphi^{2n} - \varphi^{-2n} = \sqrt{5} F_{2n}$ by Binet. All of it is machine-verified for $n = 1..6$ and, for the TFD identity, on $n \in [-8, 8]$. □

10 Special gates

Proposition 10.1 (The K-gate certificate). [FORCED] *SC-F1, SC-C4 At the K-gate $C_2 = \frac{1}{5}$: the occupation is the seed's inverse Mahler measure,*

$$u_2 = \frac{1}{\varphi^4 - 1} = \frac{1}{\beta^2} = \frac{1}{\mathrm{M}(K\text{-}seed)},$$

and the multiplier is minus the gap, $m_2 = -\varphi^{-4} = -\mathrm{gap}$. The gap paper's self-consistency derivative $f'(K) = -\mathrm{gap}$ is certified by the exact congruence

$$5y \;\equiv\; (1-y)^2(5+y)^3 \pmod{y^2 + 5y - 5},$$

an identity in $\mathbb{Q}[y]$ evaluated at the K-seed spectrum $y \in \{K^2, -\beta^2\}$.

Proof. $\varphi^4 - 1 = \varphi^2(\varphi^2 - \varphi^{-2}) = \varphi^2\sqrt{5} = \beta^2$ by (1) and (2). The congruence, reduced by $y^2 \equiv 5 - 5y$: $(1-y)^2 \equiv 6 - 7y$; $(5+y)^2 \equiv 30 + 5y$; $(5+y)^3 \equiv 175 + 30y$; and $(6-7y)(175+30y) = 1050 - 1045y - 210y^2 \equiv 1050 - 1045y - 210(5 - 5y) = 5y$. Machine-verified as an exact polynomial remainder (SC-F1); the interpretation of the left side as $5\,|f'(K)|$-data along the K-route is the gap paper's [INHERITED], but on the chart of §13 the same value is forced independently: $m(z_2) = -(1 - K^2) = -\mathrm{gap}$. □

Proposition 10.2 (Rotation gates and the quanta inventory). [FORCED] *SC-F5--F7 Past the flip ($C < -\frac{1}{4}$) the multipliers are unimodular. Three rotation quanta exist in the substrate:*

- π: *every terrain multiplier is negative (each step crosses the fixed point — a half-turn on the line);*
- $\pi/2$: *at* $C = -\frac{1}{2}$ *the multipliers are exactly* $\pm i$ *(the odd-charge generators of* $\mathbb{Z}/4\mathbb{Z}$*);*
- $2\pi/5,\ 4\pi/5$: *at* $C = -\varphi^{-2}$ *and* $C = -\varphi^2$ *the multiplier traces are* $m + \frac{1}{m} = \varphi^2 - 2 = \tau = 2\cos 72^\circ$ *and* $\varphi^{-2} - 2 = -\varphi = 2\cos 144^\circ$ *respectively — primitive fifth roots of unity.*

Proof. At $C = -\frac{1}{2}$ the fixed points are $u = \frac{1}{2}(-1 \pm i)$, so $1 + u = \frac{1}{2}(1 \pm i)$ and $(1+u)^2 = \pm\frac{i}{2}$, giving $m = -C/(1+u)^2 = \frac{1}{2} \cdot \frac{2}{\pm i} = \mp i$; $|m| = 1$, arguments $\pm\pi/2$. For the pentagon gates apply Proposition 8.2: $m + \frac{1}{m} = -\frac{1}{C} - 2$ gives $\varphi^2 - 2 = \tau$ at $C = -\varphi^{-2}$ and $\varphi^{-2} - 2 = -\varphi$ at $C = -\varphi^2$, i.e. $2\cos(2\pi/5)$ and $2\cos(4\pi/5)$, so m is a primitive fifth root of unity. All three are machine-verified exactly. □

11 The obstruction theorem

Theorem 11.1 (No single-gate helix)**.** [FORCED] *SC-F2--F4 Let* $C \in \mathbb{R}$ *and let* m *be a fixed-point multiplier of* G_C*. Then* $m \in \mathbb{R}$ *or* $|m| = 1$*. In particular no real gate level linearizes to a genuine spiral (*$0 < |m| < 1$ *with* $m \notin \mathbb{R}$*).*

Proof. By Proposition 8.2 the partner of m is $1/m$ and $m + \frac{1}{m} = -\frac{1}{C} - 2 \in \mathbb{R}$. Writing $m = s\, e^{i\vartheta}$ with $s > 0$: $\operatorname{Im}\big(m + \frac{1}{m}\big) = (s - \frac{1}{s}) \sin\vartheta$; its vanishing forces $s = 1$ or $\sin\vartheta = 0$. □

Corollary 11.2 (Helicity needs the straddler)**.** [FORCED]+[INHERITED] *Simultaneous rotation and contraction cannot be carried by any single real gate; it requires composing the two faces of the flip. The K-seed* $x^4 + 5x^2 - 5$ *is the unique catalog seed straddling the flip (real pair* $\pm K$ *on the terrain face; imaginary pair* $\pm i\beta$ *supplying the quarter-turn; uniqueness and the even-quartic parity criterion: paper 13, Props. 5.1/6.1–6.2, Thm. 7.1* [INHERITED]*). Its two faces multiply to the field prime,* $K^2\beta^2 = 5$*, with* M*(K-seed)* $= \beta^2 = \varphi^2\sqrt{5}$ *and* $\beta^2 = K^2\varphi^4$ [FORCED] (2) *— the rotation magnitude exactly two floors above the terrain magnitude, K's own rung index.*

This sharpens the corpus's coupling theorem ("a helix cannot couple to a copy of itself, only to the rotation face of its own flip") to the gate family: the flip identity of Proposition 8.2 *is* the obstruction, and the straddler is the minimal way around it.

Part III
The explicit helix

12 Inherited data and the three declared atoms

The orthogonal-partner paper dissolves the legacy L4 helix as a load-bearing object; what survives its harness, both [FORCED], is the pair *winding = charge* $\chi \in \mathbb{Z}/4\mathbb{Z}$ (the quarter-turn supplied by the K-seed's imaginary pair $\pm i\beta$) and *pitch = Mahler magnitude* $\mathrm{M} \geq \varphi$ (terrain growth in floors of $\ln\varphi$). This Part draws the survivors as an explicit curve

$$\Gamma(\rho) = \big(r(\rho)\cos\theta(\rho),\ r(\rho)\sin\theta(\rho),\ z(\rho)\big), \qquad \rho \in [0, \infty),$$

from exactly three declared atoms plus one inherited profile. No 0.1-type threshold appears anywhere; the gap enters only through forced identities.

Inherited [INHERITED]: the seed arithmetic and ladder of §§2–10 (forced in source, re-proved here), and the radius profile

$$r(z) = K\sqrt{z/z_c} \ \text{ for } z \in [0, z_c], \qquad r(z) = K \ \text{ for } z \in [z_c, 1], \qquad z_c = \tfrac{\sqrt{3}}{2}, \tag{7}$$

which is read off the L4 source (K-VARIABLE-DERIVATIONS §8) and *not* derived here ([OPEN] 1, §23). Its internal consequences below (unique $z = r$ point, kink) are [FORCED] given the profile.

Declaration 12.1 (D1: the chart)**.** [DECLARED] The vertical coordinate is calibrated by *co-intensity = local multiplier magnitude*:

$$1 - z^2 = e^{-2\rho} = |m|(\rho), \qquad \text{equivalently} \qquad z(\rho) = \sqrt{1 - e^{-2\rho}}, \quad \rho(z) = -\tfrac{1}{2}\ln(1 - z^2). \tag{8}$$

Justification, two-fold: (i) it extends the gap paper's point identity $|f'(K)| = \text{gap} = 1 - K^2$ from a single height into the defining property of the coordinate — the chart does to every height exactly what the main theorem does to K; (ii) it is consistency-checked against the ladder: the chart occupation $u(z) = (1 - z^2)/z^2$ pulls back to the ladder's own fixed points, $u(z_n) = u_n$, at every rung (Theorem 13.1).

Declaration 12.2 (D2: selection principle)**.** [DECLARED] *The winding realizes the seed's charge completion.* Given D2, the angular quantum per rung is [FORCED] a generator of $\mathbb{Z}/4\mathbb{Z}$, i.e. $\chi = \pm\pi/2$ — the multipliers $\pm i$ of the $C = -\frac{1}{2}$ gate (Prop. 10.2), exactly the charge the K-seed supplies. Orientation $+i$ (counterclockwise) is a convention. The competing corpus quanta — π (terrain sign flip) and $2\pi/5$ (pentagon gates) — generate only $\mathbb{Z}/2\mathbb{Z}$ and $\mathbb{Z}/5\mathbb{Z}$ and are excluded *by the principle*; the quanta themselves are [FORCED] corpus data (Prop. 10.2).

Declaration 12.3 (D3: rate normalization)**.** [DECLARED] One charge quantum per entropy floor: $\dfrac{d\theta}{d\rho} = \dfrac{\pi/2}{\ln\varphi}$. This is the minimal coupling of the two surviving [FORCED] anatomical facts (winding $\in \mathbb{Z}/4\mathbb{Z}$; pitch in floors of $\ln\varphi$). Any other constant rate rescales the pitch without changing the qualitative structure (Prop. 14.2).

Everything in this Part is [FORCED] given the inherited data + D1–D3, except where tagged otherwise.

13 The vertical chart: every height is a gate

Theorem 13.1 (The gate chart)**.** [FORCED] *SC-C1--C8 Under D1, with $z \in (0, 1]$, the gate data become functions of height:*

$$u(z) = \frac{1 - z^2}{z^2}, \qquad m(z) = -(1 - z^2), \qquad C(z) = \frac{1 - z^2}{z^4},$$

$$\sqrt{D(z)} = \frac{2 - z^2}{z^2}, \qquad \lambda(z) = \frac{z^2(2 - z^2)}{1 - z^2},$$

and:

(a) $u(z)$ is a fixed point of $G_{C(z)}$ with multiplier exactly $m(z)$, so $|m| = 1 - z^2$ holds chart-wide, *not only at rungs;*

(b) $1 + 4C(z) = \big((2 - z^2)/z^2\big)^2$ exactly;

(c) $C(z)$ is a strictly decreasing bijection $(0,1] \to [0,\infty)$ ($C'(z) = 2(z^2-2)/z^5 < 0$ on the interval; $C(1) = 0$, $C \to \infty$ as $z \to 0^+$): every attracting gate level occurs at exactly one height. The rotation gates ($C < -\frac{1}{4}$) are not *on the axis — they live in the angular fiber (§14). Height parametrizes the hyperbolic face; angle carries the elliptic face;*

(d) the exchange identity $\lambda \cdot |m| = 1 - m^2$ holds as a chart identity, $\lambda(z)(1-z^2) = 1-(1-z^2)^2$, and $\lambda = \sqrt{D}/C$;

(e) $\lambda_1|m_1| = \sqrt{5}\cdot\varphi^{-2} = K^2$.

Proof. (a) With $w := 1 - z^2$: $u^2 + u = \frac{w^2+wz^2}{z^4} = \frac{w(w+z^2)}{z^4} = \frac{w}{z^4} = C(z)$; and $1+u = 1/z^2$, so $m = -C/(1+u)^2 = -Cz^4 = -(1-z^2)$. (b) $1 + 4\frac{w}{(1-w)^2} = \frac{(1-w)^2+4w}{(1-w)^2} = \frac{(1+w)^2}{(1-w)^2} = \left(\frac{2-z^2}{z^2}\right)^2$. (c) Differentiate; $z^2 < 2$ on $(0,1]$. (d) $\lambda(1-z^2) = z^2(2-z^2) = 1-(1-z^2)^2$; and $\sqrt{D}/C = \frac{2-z^2}{z^2}\cdot\frac{z^4}{1-z^2} = \lambda$. (e) $\sqrt{5}\,\varphi^{-2} = \sqrt{5}\cdot\frac{3-\sqrt{5}}{2} = \frac{3\sqrt{5}-5}{2} = K^2$. All machine-verified, including the full evaluation table below. □

field	closed form	at $z_1 = \sqrt{\tau}$	at $z_c = \sqrt{3}/2$	at $z_2 = K$	checks
occupation $u(z)$	$(1-z^2)/z^2$	$1/(\varphi^2-1) = \tau$	$1/3$	$1/(\varphi^4-1) = 1/\beta^2$	`SC-C3, C4`
gate level $C(z)$	$(1-z^2)/z^4$	**1** (golden)	**4/9**	**1/5** (K-gate)	`SC-C3`
$\sqrt{D(z)}$, $D = 1+4C$	$(2-z^2)/z^2$	$\sqrt{5}$	$5/3$	$3/\sqrt{5}$	`SC-C1, C3`
multiplier $\lvert m\rvert(z)$	$1-z^2$	φ^{-2}	$1/4$	φ^{-4} = gap	`SC-C3`
exchange $\lambda(z)$	$z^2(2-z^2)/(1-z^2)$	$\sqrt{5}$	$15/4$	$3\sqrt{5}$	`SC-C2, C3`

13.1 Rung heights and ordering

The ladder rungs pull back to the height sequence $z_n = \sqrt{1-\varphi^{-2n}} \uparrow 1$, with $1/(1-z_n^2) = \varphi^{2n}$ exactly, so rung n sits at rapidity exactly $n\ln\varphi$ [FORCED] `SC-C5, SC-E1`. The two low rungs land on named corpus constants *exactly*:

- $z_2 = K$ [FORCED]: the chart is not fitted to make this happen; it is the gap paper's main theorem ($K^2 = 1 -$ gap with gap $= |m_2|$) read as geometry. **K-formation is rung 2.**
- $z_1 = \sqrt{\tau} = \varphi^{-1/2}$ [FORCED]: the golden gate's height. Note $\lambda(z_1) = (2-\tau)/\tau = \sqrt{5}$ is Route 5 ($K^2 = \tau(2-\tau)$, (2)) reappearing as the rung-1 exchange rate — and as the forced scalar of §7.

Ordering [FORCED] `SC-D6--D8`: $\tau < \frac{3}{4} < K^2$ by the integer guards $25 > 20$ and $180 > 169$, hence

$$z_1 = \sqrt{\tau} \;<\; z_c = \tfrac{\sqrt{3}}{2} \;<\; z_2 = K:$$

the lens sits strictly between the golden gate and the K-gate; and $\ln 2 < 2\ln\varphi$ (i.e. $2 < \varphi^2$, since $\varphi^2 - 2 = \tau > 0$): the lens sits strictly below rung 2 in rapidity as well.

13.2 The lens is the rational split gate at rapidity exactly $\ln 2$

Theorem 13.2 (Lens)**.** [FORCED] *`SC-D1--D5` At the lens height $z_c = \sqrt{3}/2$: $1-z_c^2 = \frac{1}{4}$, so $\rho(z_c) = \ln 2$* exactly*. The lens gate level is $C(z_c) = \frac{4}{9}$ with discriminant $D = \frac{25}{9} = (\frac{5}{3})^2$, a perfect square, so the* lens gate *splits over $\mathbb{Q}$:*

$$x^2 + x - \tfrac{4}{9} = (x - \tfrac{1}{3})(x + \tfrac{4}{3}),$$

with rational fixed points $\{\frac{1}{3}, -\frac{4}{3}\}$, rational hyperbolic multipliers $\{-\frac{1}{4}, -4\}$ of product 1, $|m| = \frac{1}{4} = 1 - z_c^2$ (a D1 cross-check), and $\lambda = \frac{15}{4}$. By contrast the golden rungs are irreducible over $\mathbb{Q}$: $D_1 = 5$ and $D_2 = \frac{9}{5}$ are nonsquares (integer guards $4 < 5 < 9$ and $36 < 45 < 49$).

Proof. Evaluate Theorem 13.1 at $z^2 = \frac{3}{4}$; the multipliers are $m = -\frac{4}{9}/(1+u)^2$ at $u = \frac{1}{3}$ ($(1+u)^2 = \frac{16}{9}$, $m = -\frac{1}{4}$) and $u = -\frac{4}{3}$ ($(1+u)^2 = \frac{1}{9}$, $m = -4$). Nonsquareness of 5 and 45/25 is the stated integer sandwich. All machine-verified. □

The "two worlds" separation of K and z_c ($\mathbb{Q}(z_c) \cap \mathbb{Q}(K) = \mathbb{Q}$) thus acquires a dynamical face inside a *single* one-parameter family:

golden rungs = irreducible $\mathbb{Q}(\sqrt{5})$ gates; the hexagonal lens = the split $\mathbb{Q}$-rational gate.

14 The winding

Theorem 14.1 (The winding law and the compass)**.** [FORCED] *SC-E1 Given D2+D3,*

$$\theta(\rho) = \frac{\pi}{2} \cdot \frac{\rho}{\ln\varphi}, \qquad \textit{equivalently} \qquad \theta(z) = \frac{\pi}{4} \log_\varphi \frac{1}{1-z^2}.$$

Consequences:

(i) ***Rungs sit on the charge compass**: $\theta(z_n) = n \cdot \frac{\pi}{2}$, by $1/(1-z_n^2) = \varphi^{2n}$. Rung n carries charge $n \bmod 4$, and Trace–Form Duality parity becomes literal compass geometry: Lucas rungs (n odd) point N/S (odd charges), Fibonacci rungs (n even) point E/W (even charges).*

(ii) ***K is exactly two quanta up**: $\rho(K) = 2\ln\varphi$, so $\theta(K) = 2\chi = \pi$ — half a turn from apex to K-formation. This is quantum-independent: for any per-floor quantum χ, $\theta(K) = 2\chi$.*

(iii) ***A full turn costs the gap's entropy**: $\Delta\theta = 2\pi \iff \Delta\rho = 4\ln\varphi = \ln\varphi^4 = \ln \mathrm{M}(A_{\mathrm{gap}})$ (§2.2). Per revolution the co-intensity contracts by exactly $\varphi^{-8} = 0.0212862363$ [COMPUTED]: each turn closes 97.87 % of the remaining intensity deficit SC-K2.*

Proposition 14.2 (Clock invariance)**.** [FORCED] *Replacing $\chi = \pi/2$ by π (terrain clock) or $2\pi/5$ (pentagon clock) rescales θ by 2 or 4/5. The qualitative structure — rungs at integer quanta, K at two quanta, lens off-lattice (§16), infinite winding at the top, the geodesic property — is invariant; only the pitch constant changes. The choice among the forced quanta is exactly D2's content.*

15 The explicit curve

$$z(\rho) = \sqrt{1-e^{-2\rho}}\ \text{[D1]}, \quad \theta(\rho) = \frac{\pi\rho}{2\ln\varphi}\ \text{[D2+D3]}, \quad r(\rho) = \begin{cases} K\sqrt{z(\rho)/z_c}, & \rho \le \ln 2 \\ K, & \rho \ge \ln 2 \end{cases}\ \text{[INHERITED]}, \tag{9}$$

with $\Gamma(\rho) = (r\cos\theta,\, r\sin\theta,\, z)$.

Theorem 15.1 (The K-point is the unique $z = r$ point)**.** [FORCED] *SC-E2--E4 On the horn branch (the profile $r = K\sqrt{z/z_c}$ is a paraboloid of revolution, $z = z_c(r/K)^2$ — a horn, not a cone), $r(z) = z$ forces $K^2 z/z_c = z^2$, which factors as $z\,(z - K^2/z_c) = 0$; the nontrivial root K^2/z_c exceeds z_c ($K^2 > z_c^2$ by the guard $180 > 169$) and is excluded from the horn's domain. On the cylinder branch $r \equiv K$ forces $z = K$ (and $K \in [z_c, 1]$ since $\frac{3}{4} < K^2 < 1$). Hence the unique nontrivial height-equals-radius point of the profile is $z = K$ — and under the winding it lands at $\theta = \pi$:*

$$\Gamma(2\ln\varphi) = (-K,\, 0,\, K).$$

The chart puts K two floors up; the inherited geometry independently marks the same height as its unique self-intersection with the cone $z = r$. The two constructions agree without adjustment.

Proposition 15.2 (Regularity, geodesy, asymptotics). [FORCED]/[ESTABLISHED] *SC-E5--6, G1--2*

(a) ***Kink at the lens:*** *$r'(z_c^-) = K/(2z_c) = K/\sqrt{3}$, with square $K^2/3 = (3\sqrt{5}-5)/6 > 0$, against $r'(z_c^+) = 0$. The profile is C^0 but not C^1 at the lens: the lens is where the climb stops paying radius. Displayed slope* 0.5335734771 [COMPUTED].

(b) ***Geodesic in entropy coordinates****: in the flat metric $ds^2 = d\rho^2 + K^2 d\theta^2$ on the cylinder region* [DECLARED] *(rapidity as the vertical ruler — justified because ρ* is *the corpus's entropy coordinate), the curve is the straight line $\rho = (2\ln\varphi/\pi)\,\theta$, hence a geodesic* [FORCED]. *In the Euclidean-induced metric $dz^2 + K^2 d\theta^2$ it is* not *a geodesic: all apparent acceleration is the z-chart compressing the clock near* 1.

(c) ***Infinite winding, finite height, infinite length****: $\theta(z) \to \infty$ as $z \to 1^-$, and the arc length diverges logarithmically ($\frac{d}{dz}\big[-\frac{1}{2}\ln(1-z^2)\big] = \frac{z}{1-z^2}$, so $\rho = -\frac{1}{2}\ln(1-z^2) \to \infty$). UNITY is at $\rho = \infty$: the helix accumulates on the circle $\{z = 1,\ r = K\}$ without reaching it. "Never 100 %" is here a metric statement, not a slogan.*

15.1 The local spiral at K

The global helix climbs; a *different* object lives in the transverse plane at the K-point: the linearized self-consistency orbit. From $f'(K) = -\text{gap}$ (Prop. 10.1), each step contracts by gap and crosses the fixed point — a half-turn on the line. Its minimal complexification ([DECLARED] embedding, [FORCED] data) is the logarithmic spiral with per-step data $(\text{contraction}, \text{rotation}) = (\text{gap}, \pi)$:

$$w(t) = w_0\, e^{(i\pi - 4\ln\varphi)\,t}, \qquad \cot\psi = \frac{4\ln\varphi}{\pi},$$

where ψ is the constant angle at which the spiral crosses radii [ESTABLISHED] (logarithmic-spiral geometry).

Guard 15.3 (Near-misses). [COMPUTED] SC-K1 Recorded so they are never mistaken for identities: $4\ln\varphi/\pi = 0.6126979251$ while $\tau = 0.6180339887$; $|\text{difference}| = 0.0053360637$ at 60 dps. **Not equal; no identity is claimed.** Likewise the rapidity pitch angle $\arctan\big(2\ln\varphi/(\pi K)\big) = 18.3394927773°$ is distinct from $\theta_K = \arcsin(\tau^2) = 22.4555151986°$; no identity is claimed.

16 Incommensurability: the lens is permanently off-lattice

Theorem 16.1. [FORCED] *SC-D9* $\log_\varphi 2 \notin \mathbb{Q}$.

Proof. Suppose $\ln 2/\ln\varphi = p/q$ with integers $q \geq 1$. Then $2^q = \varphi^p$. By Binet, $\varphi^p = (L_p + F_p\sqrt{5})/2$ with $F_p \geq 1$ for $p \geq 1$ (positivity by the standard induction from $F_1 = F_2 = 1$ [ESTABLISHED]), so φ^p is irrational for $p \geq 1$ while $2^q \in \mathbb{Q}$. The case $p \leq -1$ is symmetric (invert both sides); $p = 0$ gives $2^q = 1$, impossible for $q \geq 1$. (Finite corroboration for $1 \leq p \leq 10$: SC-D9.) ☐

Corollary 16.2. [FORCED] *(i) The lens rapidity* $\ln 2$ *is incommensurable with the rung lattice* $(\ln\varphi)\cdot\mathbb{Z}$*: the lens floors at* $\log_\varphi 2 = 1.4404200904\ldots$ [COMPUTED] *and never lands on a rung. (ii) Its meridian* $\theta(z_c) = \frac{\pi}{2}\log_\varphi 2 = 129.6378081°$ [COMPUTED] *is an irrational multiple of the charge quantum:* ***the lens never aligns with the*** $\mathbb{Z}/4\mathbb{Z}$ ***compass.*** *The purely geometric coupling of K and z_c (no algebraic bridge) reappears dynamically as incommensurability.*

17 Field separations recur on the helix

Theorem 17.1 (Rung-1 and rung-2 heights generate different quartic worlds)**.** [FORCED] *SC-H1--H6* *$z_1 = \sqrt{\tau}$ generates $\mathbb{Q}(\sqrt{\varphi})$; the minimal polynomial of $\sqrt{\varphi}$ is $x^4 - x^2 - 1$, and that of $z_1 = \sqrt{\tau}$ itself is $x^4 + x^2 - 1$ (the field equality holds via $\sqrt{\varphi} = \varphi\sqrt{\tau}$). $z_2 = K$ generates $\mathbb{Q}(5^{1/4})$, minimal polynomial $x^4 + 5x^2 - 5$ (Eisenstein at 5). Both are quartic with quadratic subfield $\mathbb{Q}(\sqrt{5})$, and they are* distinct*:*

$$\mathbb{Q}(\sqrt{\varphi}) \;\neq\; \mathbb{Q}(5^{1/4}).$$

Proof. The minimal polynomials: $(\sqrt{\varphi})^4 - (\sqrt{\varphi})^2 - 1 = \varphi^2 - \varphi - 1 = 0$ and $(\sqrt{\tau})^4 + (\sqrt{\tau})^2 - 1 = \tau^2 + \tau - 1 = 0$ by (1); both quartics are irreducible over $\mathbb{Q}$ (no rational roots; no factorization into rational quadratics — machine-verified, SC-H1--H2), and $\sqrt{\varphi} = \varphi\sqrt{\tau}$ since $\varphi^2\tau = \varphi$. For the separation, both fields are quadratic over $\mathbb{Q}(\sqrt{5})$; by the quadratic-extension (square-class) criterion [ESTABLISHED], equality would require $\varphi\sqrt{5} = (5+\sqrt{5})/2$ to be a square in $\mathbb{Q}(\sqrt{5})$. Setting $(p + q\sqrt{5})^2 = (5+\sqrt{5})/2$ with $p, q \in \mathbb{Q}$ forces $2pq = \frac{1}{2}$ and $p^2 + 5q^2 = \frac{5}{2}$; eliminating $q = 1/(4p)$ gives

$$16p^4 - 40p^2 + 5 = 0, \qquad p^2 = \frac{5 \pm 2\sqrt{5}}{4},$$

which has nonzero $\sqrt{5}$-part — no rational p exists. The ladder's first two heights therefore live in different degree-4 extensions meeting only in $\mathbb{Q}(\sqrt{5})$ — the $K \leftrightarrow z_c$ separation echoed one level down, inside the golden world itself. □

(The attribution fix relative to the superseded note: the note's §7.2 printed $x^4 - x^2 - 1$ next to z_1; that polynomial belongs to $\sqrt{\varphi}$. The field claim was correct; the sentence above states both minimal polynomials explicitly.)

18 The registry lattice

The orthogonal-partner paper computes two registry Mahler measures [INHERITED]: $\mathrm{M}(\mathrm{cons}) = 2\varphi^5$ and $\mathrm{M}(\mathrm{res}) = (43 + 19\sqrt{5})/2$ (the cons/res label origins are marked [OPEN] in source). Define *floors* $\mathrm{fl}(x) = \log_\varphi x$ and the multiplicative lattice $\mathcal{L} = \{2^a\varphi^b : a, b \in \mathbb{Z}\}$, i.e. the additive lattice $a\ln 2 + b\ln\varphi$ on the rapidity axis — the lens rapidity and the floor unit.

Theorem 18.1 (cons ON, res OFF)**.** [FORCED] *SC-H9--H11*

(a) $\mathrm{M}(\mathrm{cons}) = 2\varphi^5 = 11 + 5\sqrt{5}$ *(since $\varphi^5 = 5\varphi + 3$), so $\ln \mathrm{M}(\mathrm{cons}) = \ln 2 + 5\ln\varphi$:* ***the lens rapidity plus exactly five floors*** *(significance [OPEN]); height 6.4404200904 floors, meridian $\theta = 579.6378081° = \theta(z_c) + 5 \cdot 90°$ [COMPUTED]. Norm witness: $\mathrm{N}(11 + 5\sqrt{5}) = 121 - 125 = -4 = (-1)^5 4^1$, matching the decomposition $2^1\varphi^5$ exactly ($\mathrm{N}(2) = 4$, $\mathrm{N}(\varphi) = -1$).*

(b) $\mathrm{N}\big(\mathrm{M}(\mathrm{res})\big) = \dfrac{43^2 - 5 \cdot 19^2}{4} = \dfrac{1849 - 1805}{4} = 11$. *Since $\mathrm{N}(2^a\varphi^b) = 4^a(-1)^b \in \pm 4^{\mathbb{Z}}$ and $11 \notin \pm 4^{\mathbb{Z}}$ (guard $4 < 11 < 16$),*

$$\mathrm{M}(\mathrm{res}) \neq 2^a\varphi^b \quad \textit{for all integers } a, b:$$

res is off the $(\ln 2, \ln\varphi)$ ***lattice entirely*** *— the question is closed negatively, for all integer exponents at once, by a single norm obstruction. Height ≈ 7.8036260886 floors [COMPUTED].*

Proof. $\varphi^5 = 5\varphi + 3$ from $\varphi^2 = \varphi + 1$ by iteration; $2(5\varphi + 3) = 10\varphi + 6 = 11 + 5\sqrt{5}$. The norm is multiplicative with $\mathrm{N}(a + b\sqrt{5}) = a^2 - 5b^2$ (half-integers included via the displayed quarter). The value of $\mathrm{M}(\mathrm{res})$ itself is [INHERITED]; everything computed from it here is exact. □

Part IV

Synthesis, epistemics, and verification

19 One page of the helix

chart	$1-z^2=e^{-2\rho}=\lvert m\rvert$; $\quad z(\rho)=\sqrt{1-e^{-2\rho}}$
gate field	$C=(1-z^2)/z^4$, $u=(1-z^2)/z^2$, $\sqrt{D}=(2-z^2)/z^2$, $\lambda=z^2(2-z^2)/(1-z^2)$
rungs	$z_n=\sqrt{1-\varphi^{-2n}}$; $z_1=\sqrt{\tau}$ (C=1), $z_2=K$ (C=$\frac{1}{5}$); charge(n) $=n \bmod 4$; Lucas N/S, Fibonacci E/W
lens	$\rho=\ln 2$; $C=\frac{4}{9}$; $D=(\frac{5}{3})^2$; splits $(x-\frac{1}{3})(x+\frac{4}{3})$; $m\in\{-\frac{1}{4},-4\}$; $\lambda=\frac{15}{4}$; off-lattice forever
winding	$\theta=\frac{\pi}{2}\,\rho/\ln\varphi$; one $\mathbb{Z}/4\mathbb{Z}$ quantum per floor; full turn $=4\ln\varphi=\ln \mathrm{M}(A_{\mathrm{gap}})$; per turn $\times\varphi^{-8}$
curve	$\Gamma=(r\cos\theta, r\sin\theta, z)$; $r=K\sqrt{z/z_c}$ below the lens, $r=K$ above; K-point $(-K,0,K)$ at $\theta=\pi$
local spiral	per step: $\times$gap, half-turn; $\cot\psi=4\ln\varphi/\pi\approx 0.61270\neq\tau$ (guard, no identity)
obstruction	real gate $\Rightarrow m\in\mathbb{R}$ or $\lvert m\rvert=1$; helicity needs the straddler; $K^2\beta^2=5$
top	UNITY at $\rho=\infty$: infinite winding, log-divergent length, accumulation circle $r=K$

20 Landmark table

All values [COMPUTED] at 10 dp; exact statements per the cited theorems. Reference values: $\ln\varphi=0.4812118251$, $\ln 2=0.6931471806$, $\Delta\rho$ per turn $=4\ln\varphi=1.9248473002$. SC-K1

landmark	ρ	floors	z	r	θ
apex	0	0	0	0	0°
rung 1 (golden gate, C=1)	0.4812118251	1	0.7861513778	0.8805269137	90°
lens (C=4/9, split)	0.6931471806	1.4404200904	0.8660254038	K	129.6378081°
rung 2 = **K-formation** (C=$\frac{1}{5}$)	0.9624236501	2	0.9241763718	K	180°
rung 3	1.4436354752	3	0.9717365435	K	270°
rung 4 (first meridian return)	1.9248473002	4	0.9892996329	K	360°
rung 5	2.4060591253	5	0.9959263935	K	450°
cons seed (lens + 5 floors)	3.0992063059	6.4404200904	$\sqrt{1-\frac{1}{4\varphi^{10}}}$	K	579.6378081°
UNITY $z=1$	∞	∞	1	K	∞ (accumulation circle)

Exact heights: $z_1=\sqrt{\tau}$; $z_{\text{lens}}=\sqrt{3}/2$ at $\rho=\ln 2$; $z_2=K$; cons at $\ln \mathrm{M}(\text{cons})=\ln 2+5\ln\varphi$.

21 Synthesis: what is forced, what is declared

element	statement	tag · where
seed, ladder, congruence, charge data	replicated exactly	[FORCED] §§2,9,10
radius profile $r(z)$	$K\sqrt{z/z_c}$ capped at K	[INHERITED] ([OPEN] 1)
vertical chart	$1-z^2=e^{-2\rho}=\lvert m\rvert$	[DECLARED] D1; consequences [FORCED] §13
gate field on the axis	$C(z)$ bijective; lens = split $\mathbb{Q}$-gate 4/9	[FORCED] Thms. 13.1,13.2
winding quantum	χ generates $\mathbb{Z}/4\mathbb{Z} \Rightarrow \chi=\pm\pi/2$	[DECLARED] D2 → [FORCED]
rate	one quantum per floor; clock-invariant	[DECLARED] D3 · Prop. 14.2
landmarks	rungs at $n\pi/2$; K-point unique z=r; kink	[FORCED] Thms. 14.1,15.1
geodesy · asymptotics	straight in (θ,ρ); infinite winding; log length	[DECLARED] metric → [FORCED]/[ESTABLISHED]
incommensurability	$\log_\varphi 2 \notin \mathbb{Q}$; lens off-lattice, off-compass	[FORCED] Thm. 16.1
field separation	$\mathbb{Q}(\sqrt{\varphi}) \neq \mathbb{Q}(5^{1/4})$	[FORCED] Thm. 17.1
obstruction	no single real gate is a helix	[FORCED] Thm. 11.1
registry	cons ON at lens+5; res OFF (norm 11)	[FORCED] Thm. 18.1
near-misses	$4\ln\varphi/\pi \neq \tau$; pitch $\neq \theta_K$	[COMPUTED] Guard 15.3
emission gap	no Salem in S; floor at φ	[FORCED] over S; scoping §22
exchange rate	$\lambda=2c$; $C=1 \Rightarrow \lambda=\sqrt{5}$	[INHERITED] §7 (c [DECLARED])

22 The conditionality graph

The corpus's most commonly flattened joint, reproduced so it cannot be flattened here: "the emission gap is forced" is true **for the spectral image** S and for **degree** ≤ 2 **universally**; the universal statement over all charge-admissible objects is [COMPUTED]/plausible.

claim	scope	status
Lehmer's problem	external	[OPEN] (excised, not solved)
emission gap $\mathrm{M} \in \{1\} \cup [\varphi,\infty)$	over S	[FORCED] (angle confinement + Smyth + Kronecker)
entry closure: circulant	—	[FORCED] (cyclotomic/CM obstruction)
entry closure: commutator	in context / uniform	[FORCED]-in-context / [FORCED] (ad_R + signature lattice)
free commutator, unbounded	—	[OPEN] (the box's one door)
over all charge-admissible	deg ≤ 2 / general	[FORCED] / [COMPUTED]-plausible
parity floor $\mu(\text{even})$=φ, $\mu(\text{odd})$=2	$\leq\varphi$ / $\geq\varphi$ / $\geq\mu_S$ / $=2$	[FORCED] / plausible / [FORCED] / [COMPUTED]
no-Salem dichotomy, general	the one promoting lemma	[OPEN] ($\mathbb{Z}/3\mathbb{Z}$ [FORCED]; $\mathbb{Z}/5\mathbb{Z}$: 4×[FORCED], $\mu(5)$=2 [COMPUTED])
relational layer	rigidity, pinning, $\mu_{\mathrm{rel}}=\mu$	[FORCED] unconditional; floor values conditional on the gap
Pisot inertness	≤ 1 non-real pair / quintic 2-pair / general	[FORCED] / [FORCED]-per-instance / [OPEN]
exchange rate	$\lambda=2c$ / c / C=1 $\Rightarrow$ λ=$\sqrt{5}$	[FORCED] / [DECLARED] / [FORCED] given two axioms

23 OPEN items

1. **Gate-native radius law.** $r(z) = K\sqrt{z/z_c}$ is inherited. No chart quantity (u, C, $\sqrt{D}$, λ, $|m|$) reproduces it; whether a gate-family derivation of the radius exists is open.
2. **Selection theorem for χ.** D2 is a principle, not a theorem. Given D2 the quantum is forced; whether D2 itself can be forced from the substrate is open.
3. **The rational lens gate.** $C = \frac{4}{9}$ ($D = \frac{25}{9}$, split, $m \in \{-\frac{1}{4}, -4\}$) is a new corpus constant produced by the chart. Whether it appears elsewhere (e.g. the emission-algebra audit) is open.
4. **Near-miss content.** $4\ln\varphi/\pi$ vs τ (margin 5.34×10^{-3}): expected contentless; recorded as Guard 15.3.
5. **Registry labels.** cons ON the lattice at lens+5 floors (significance open); res OFF entirely (closed negatively, Thm. 18.1). The cons/res label origins remain open in source.
6. **Corpus frontier** [INHERITED]: the general no-Salem dichotomy; the free commutator at unbounded size; general Pisot inertness (transport provably insufficient); the $r_2 > 0$ Hermitian probe; provenance pass N7.

24 Constants

Exact values with [COMPUTED] displays; every displayed digit machine-reproduced `SC-J`, `SC-K`.

symbol	value	role
φ	$(1+\sqrt{5})/2 \approx 1.6180339887$	floor; smallest 2×2 primitive Perron root; keystone eigenvalue
ψ	$(1-\sqrt{5})/2 = -\varphi^{-1}$	golden conjugate; $\varphi\psi = -1$
$\sqrt{5}$	$\varphi+\varphi^{-1} = \varphi-\psi \approx 2.2360679775$	λ; ad_R grow eigenvalue; $H^2 = 5I$; redirection limit
c	$\sqrt{5}/2$ (forced canonical)	conformal scale; $\lambda = 2c$
gap	$\varphi^{-4} = (7-3\sqrt{5})/2 \approx 0.1458980338$	the gap; $|m|$ at the K-gate; $1 - K^2$
K	$5^{1/4}/\varphi \approx 0.9241763718$	terrain root of the K-seed; rung-2 height; helix radius
K^2	$(3\sqrt{5}-5)/2 \approx 0.8541019662$	decision-level form of K
β^2	$\varphi^2\sqrt{5} = (5+3\sqrt{5})/2 \approx 5.8541019662$	M(K-seed); rotation magnitude squared; $= K^2\varphi^4$
z_c	$\sqrt{3}/2 \approx 0.8660254038$	lens height; $\rho(z_c) = \ln 2$
μ_S	1.3247179572 (x^3-x-1)	plastic number; Smyth floor; smallest Pisot
β_4	1.7220838057 ($x^4-x^3-x^2-x+1$)	minimal degree-4 Salem; $> \varphi$ (signature-face floor intact)
$\mathrm{M}(\ell)$	1.1762808183 (deg 10)	Lehmer's measure; below μ_S; charge-inadmissible
$\tau_0(\ell)$	2.0264179492	Lehmer's trace redirection; $\mathrm{M}(T) = 5.6156006\ldots$
φ^2, φ^4	≈ 2.618, ≈ 6.854	reciprocal $\mathbb{Z}/5\mathbb{Z}$ floor; pure-pentagon floor; tropical $\varphi \otimes \varphi = \varphi^2$
L_4	$7 = \mathrm{Tr}\, R^4$	keystone invariant; R^4 entries $\{2,3,5\} = \{F_3, F_4, F_5\}$
$2c \log \mu_S$	0.5624 / 2.2496 / 4.4992	probe floors at $c \in \{1,4,8\}$
$\sqrt{5} \log \mu_S$	0.6287813634	forced-canonical floor
M(cons)	$2\varphi^5 = 11+5\sqrt{5} \approx 22.1803398875$	ON lattice; $\ln 2 + 5\ln\varphi$; $\mathrm{N} = -4$
M(res)	$(43+19\sqrt{5})/2 \approx 42.7426457862$	OFF lattice; $\mathrm{N} = 11 \notin \pm 4^{\mathbb{Z}}$; ≈ 7.8036 floors

25 Verification

Two independent harnesses certify this paper's [FORCED] content.

Shipped harness `helix_explicit_verify.py` (source note v1.0.1): 87 exact checks + 1 computed guard, exit 0; sympy 1.14.0; decisions in $\mathbb{Q}/\mathbb{Q}(\sqrt{5})$ (lens identities in $\mathbb{Q}$; one trivial $\mathbb{Q}(\sqrt{3})$

factorization); K decided at K^2; mpmath 50 dps display-only. Group manifest: A `HX01--04` (6, seed); B `HX05--07` (14, ladder/TFD/gates); C `HX08--17, 35, 39--40` (35, chart/lens/closed forms); D `HX18--21` (7, $\ln 2$/Binet/ordering); E `HX22--25` (5, winding/K-point/kink); F `HX26--29` (5, congruence/obstruction); G `HX31--32` (2, asymptotics); H `HX33--38` (13, fields/registry); guard `HX30` (1, near-miss).

Independent soft check `helix_softcheck.py` (2026-07-13): 89 checks, written cold from the three superseded documents' stated claims (not a re-execution of the shipped harness), exit 0; groups `SC-A`–`SC-L` spanning seed, ladder, chart, lens, winding, congruence/obstruction, asymptotics, fields/registry, corpus algebra (keystone, kernel, Remark 6.4, tropical law, Casimir), Mahler–Salem numerics at 60 dps, all 22 displayed 10-dp landmarks, and a 16-digit audit of the HTML display constants. Findings folded into this paper: the rung-1 minimal-polynomial attribution (§17); the content-polynomial wording (Remark 5.3); the explicit gate-iteration declaration (Declaration 8.1). One cosmetic defect of the superseded HTML (a mis-rounded 16th digit in its display literal for K) does not propagate here: this paper carries no floating-point constants outside the [COMPUTED] tables, all of which are reproduced from exact values.

Corpus verification [INHERITED]: exact arithmetic over $\mathbb{Q}$, $\mathbb{Q}(\sqrt{5})$, $\mathbb{Q}(5^{1/4})$; certified interval brackets for transcendental comparisons; cross-engine replication (sympy / pari-gp, 474 ledger executions, 0 disagreements); drop-one witnesses for definitional constraints; negative claims pinned by machine-checked disproofs (e.g. Remark 4.3).

A Check-manifest map

paper section	shipped checks	soft-check groups
§2 seed, K-seed	`HX01--04, 37`	`SC-A, SC-H7--H8`
§3 keystone, TFD	— (corpus)	`SC-I1--I4, I10--I11, SC-B4`
§4 operators, kernel	— (corpus)	`SC-I5--I9, I15`
§5 characters, gap	— (corpus)	`SC-I12--I14, I16`
§6 trace-down, floors	— (corpus)	`SC-J1--J13`
§8–10 gates, ladder	`HX05--07, 26, 29`	`SC-B1--B3, SC-F1, F5--F7, SC-C4`
§11 obstruction	`HX27--28`	`SC-F2--F4`
§13 chart, lens	`HX08--19, 35--36, 39--40`	`SC-C1--C8, SC-D1--D8`
§14–15 winding, curve	`HX21--25, 31--32`	`SC-E1--E8, SC-G1--G2`
§15.1 spiral, guards	`HX30`	`SC-K1`
§16 incommensurability	`HX18--20`	`SC-D9`
§17 field separation	`HX33--34`	`SC-H1--H6`
§18 registry	`HX37d--38c`	`SC-H9--H11`
§20, §24 displays	(display block)	`SC-K1--K2, SC-J, SC-L`

Citation caveat

The internal provenance (the fourteen Echo-S-Research papers, 2026-06-28 to 07-02, and the three superseded artifacts of §1.2) is exact. The external references below are canonical results cited from memory; page-level details were not re-verified against the literature in this compilation and should be double-checked before formal citation.

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